

Himani Sinhmar

Postdoctoral Research Associate, Princeton University

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POSITIONS HELD **Princeton University**

Postdoctoral Research Associate, Leonard Lab

Sept. 2024–present

Advisor: Prof. Naomi Ehrich Leonard

EDUCATION **Cornell University**

Ph.D., Aerospace Engineering

Aug. 2019–Aug. 2024

Specialization in Robotics; Minor in Computer Science

Doctoral Advisor: Prof. Hadas Kress-Gazit

Indian Institute of Technology (IIT Bombay)

B.Tech. and M.Tech., Aerospace Engineering

2014–2019

Specialization in Systems and Control; Minor in Physics

Thesis Advisor: Prof. Srikant Sukumar

RESEARCH INTERESTS

My vision is to develop theoretically grounded and resource aware autonomy architectures that make robots safe, goal-directed, smoothly adaptive, and interpretable *by design*. My current research asks how can a robot even decide on a feasible action when the information it needs emerges only after it begins to act? I *design, analyze, and certify* new decision-making and control methods for interactive autonomy and *validate* them on robotic hardware. My work draws on insights from dynamical systems, control theory, neuroscience, and collective behavior.

MANUSCRIPTS IN PREPARATION

*Advised students are marked with [†] and equal contributions with *, when applicable.*

- [P1] **Himani Sinhmar**, Vaibhav Srivastava, and Naomi Ehrich Leonard. “Neuromorphic Decision Dynamics for Deadlock-free Safe Navigation in Congested Environments” (in prep.)
- [P2] **Himani Sinhmar**^{*}, Antoun K. Salib^{*,†}, and Naomi Ehrich Leonard. “Real-Time Inference of Cooperation for Path-Preserving Multi-Robot Social Navigation in Crowded Spaces.” (in prep.)
- [P3] **Himani Sinhmar**, Giovanna Amorim[†], Vaibhav Srivastava, and Naomi Ehrich Leonard. “Decentralized Neuromorphic Supervisory Control for Parsimonious Multi-Agent Persistent Environment Monitoring.” (in prep.)

JOURNAL PUBLICATIONS

- [J1] Marcus Greiff, **Himani Sinhmar**, Andreas Weiss, Kristofer Berntorp, and Stefano Di Cairano. “[Invariant Set Planning for Quadrotors: Design, Analysis, and Experiments.](#)” *IEEE Transactions on Control Systems Technology*, 33(2):449-462, 2024.
- [J2] Qingkun Liu^{*}, Wei Wang^{*}, **Himani Sinhmar**, Itay Griniasty, Jason Z. Kim, Jacob T. Pelster, Parag Chaudhari, Michael F. Reynolds, Michael C. Cao, David A. Muller, Alyssa B. Apsel, Nicholas L. Abbott, Hadas Kress-Gazit, Paul L. McEuen, and Itai Cohen. “[Microscopic, Continuum, Compliant, and Electronically Configurable Metasheet Robots.](#)” *Nature Materials*, 2024.
- [J3] **Himani Sinhmar** and Hadas Kress-Gazit. “[Guaranteed Encapsulation of Targets with Unknown Motion by a Minimalist Robotic Swarm.](#)” *IEEE Transactions on Robotics*, 40:816-830, 2023.
- [J4] Pallavi Sinha, Srikant Sukumar, and **Himani Sinhmar**. “[Consensus of Networked Double Integrator Systems under Sensor Bias.](#)” *International Journal of Adaptive Control and Signal Processing*, 37(1):278-297, 2022.

CONFERENCE
PUBLICATIONS

- [C1] **Himani Sinhmar**, Vaibhav Srivastava, and Naomi Ehrich Leonard. “[Neuromorphic Realization of Best Response in Finite-Action Games.](#)” To Appear in 2026 IEEE Conference on Decision and Control
- [C2] Ian Xul Belaustegui, **Himani Sinhmar**, Ling-Wei Kong, Andrew Michael Hein, and Naomi Ehrich Leonard. “[A Continuous-Time and State-Space Relaxation of the Linear Threshold Model with Nonlinear Opinion Dynamics.](#)” To Appear in 2026 IEEE Conference on Decision and Control
- [C3] Donggeon David Oh, Justin Lidard, Haimin Hu, **Himani Sinhmar**, Elle Lazarski, Deepak Gopinath, Emily S. Sumner, Jonathan A. DeCastro, Guy Rosman, Naomi Ehrich Leonard, and Jaime Fernández Fisac. “[Safety with Agency: Human-Centered Safety Filter with Application to AI-Assisted Motorsports.](#)” *Robotics: Science and Systems*, 2025.
- [C4] **Himani Sinhmar** and Hadas Kress-Gazit. “[Multi-Source Encapsulation With Guaranteed Convergence Using Minimalist Robots.](#)” *Distributed Autonomous Robotic Systems*, 2024.
- [C5] **Himani Sinhmar**, Marcus Greiff, and Stefano Di Cairano. “[Practical and Safe Navigation Function Based Motion Planning of UAVs.](#)” *IEEE International Conference on Robotics and Automation*, 2024.
- [C6] **Himani Sinhmar** and Hadas Kress-Gazit. “[Decentralized Control of Minimalistic Robotic Swarms for Guaranteed Target Encapsulation.](#)” *IEEE/RSJ International Conference on Intelligent Robots and Systems*, 2022.
- [C7] **Himani Sinhmar** and Srikanth Sukumar. “[Distributed Model Independent Algorithm for Spacecraft Synchronization under Relative Measurement Bias.](#)” *5th CEAS Conference on Guidance, Navigation and Control*, 2019.
- [C8] **Himani Sinhmar** and Vinod Kumar. “[Relative Autonomous Navigation Without Communication Between Spacecraft Using Line of Sight Measurements.](#)” *8th IEEE/CSAA Guidance, Navigation and Control Conference*, 2018.

WORKSHOP
PAPERS

- [W1] **Himani Sinhmar**, Itay Griniasty, Qingkun Liu, Wei Wang, Itai Cohen, and Hadas Kress-Gazit. “Design and Control of Microscopic Robot Sheet.” *ICRA Workshop*, 2024.
- [W2] **Himani Sinhmar**, Vaibhav Srivastava, Naomi Ehrich Leonard. “Neuromorphic Decision Dynamics and Control for Provably Safe and Deadlock-Free Multi-Agent Navigation in Congested Environments.” *IROS Workshop*, 2026.
- [W3] **Himani Sinhmar**, Antoun Salib, Naomi Ehrich Leonard. “Cooperation-Aware Neuromorphic Decision Dynamics for Path-Preserving Navigation in Heterogeneous Crowds.” *IROS Workshop*, 2026.

PATENTS AND
DISSERTATION

- [PT1] Marcus Greiff, Stefano Di Cairano, and **Himani Sinhmar**. “[Motion Planning and Control with Multi-Stage Construction of Invariant Sets.](#)” US Patent Application No. 18/589,946, 2025.
- [D1] **Himani Sinhmar**. “Control With Guarantees for Minimalist Robotic Swarms.” Ph.D. dissertation, Cornell University, 2024.

SELECTED
HONORS AND
AWARDS

Rising Star in Mechanical Engineering	Stanford University, 2026
Future Values Fellowship	Princeton University, 2026
Robotics Rising Star	Princeton University, 2026
Future Leader in Robotics and AI	University of Maryland, 2025
Hibbitt Postdoctoral Fellowship	Brown University, offered and declined, 2024
Caltech EAS Trailblazers Award	Caltech, 2024
Robotics: Science and Systems Pioneers	RSS Conference, 2024
Graduate Resident Fellowship	Cornell University, 2022–2024
Institute Undergraduate Research Award	IIT Bombay, 2018

PRESENTATIONS
AND POSTERS

Invited Seminars and Talks

Nov. 2025	Invited Talk at SIAM NNP Mini-Symposium on “Collective Behavior for Biologically Inspired Social Systems”	Penn State University
Sept. 2025	Invited participant, NSF visioning workshop “Formalization of Human Factors into Computational Models,” sponsored by the Mind, Machine, and Motor Nexus program of NSF.	Alexandria, VA
Apr. 2025	Invited seminar, Future leaders in robotics and AI, “Provably Scalable and Decentralized Control Design for Minimalist Robot Swarm” [Recording]	University of Maryland
Nov. 2025	Invited Trailblazers seminar for Department of CMS	Caltech
May 2024	Leonard Lab, <i>Princeton University</i> .	Princeton, NJ
Feb. 2024	GRASP Spring Seminar, <i>University of Pennsylvania</i> .	Philadelphia, PA
Feb. 2024	Automatic Coordination of Teams Lab, <i>Brown University</i> .	Providence, RI
Jan. 2024	Biorobotics Lab, <i>Carnegie Mellon University</i> .	Pittsburgh, PA
Jan. 2024	Distributed Robotics Lab, <i>MIT CSAIL</i> .	Cambridge, MA
Jan. 2023	Collective Embodied Intelligence Lab, <i>Cornell University</i> .	Ithaca, NY
May 2022	Guest lecture, Introduction to Computational Science and Engineering.	Cornell University

Conference and Poster Presentations

2026	Invited poster, Learning Dynamics in Natural and Artificial Intelligence .	Harvard University
2026	Invited poster, Diverse Intelligences Summit .	Princeton University
2026	Poster, <i>Northeast Robotics Colloquium</i> .	Princeton University
2025	Poster, <i>Northeast Robotics Colloquium</i> .	Cornell University
2024	IEEE RAS Summer School on Multi-Robot Systems	Prague
2024	Poster, <i>Robotics, Science and Systems Conference (RSS)</i> .	Delft University of Technology
2023	Poster, <i>International Conference on Robotics and Automation (ICRA)</i> .	Japan
2023	Poster, <i>Northeast Robotics Colloquium</i> .	Yale University

MENTORING AND PROFESSIONAL DEVELOPMENT	2025–2026	Antoun K. Salib, Senior Thesis Advisor , Department of Electrical and Computer Engineering, Princeton University. Thesis: <i>Path-Preserving Multi-Robot Navigation in Crowded Social Spaces with Online Cooperation Inference</i> . Sigma Xi Honor Society Book Award for Outstanding Research
	2026–2027	Rebecca Shin, Senior Thesis Advisor , Department of Mechanical and Aerospace Engineering, Princeton University. Thesis: <i>Integrated Planning and Navigation for Autonomous Agricultural Robots in Dynamic Environments</i> .
	2026–2027	Edric Zhang, Senior Thesis Advisor , Department of Electrical and Computer Engineering
	2025	Facilitating Mentor Training, CIMER Academic Mentor Training Workshop.
TEACHING EXPERIENCE	2021–2023	Autonomous Mobile Robots, Teaching Assistant, Cornell University.
	2019	Dynamics and Control, Teaching Assistant, IIT Bombay.
	2018	Spaceflight Mechanics, Teaching Assistant, IIT Bombay.
PROFESSIONAL SERVICE	Conference and Symposium Organization	
	2026	Organizing Committee, Princeton Robotics Symposium.
	2025	Website and Publicity Chair, Robotics: Science and Systems Pioneers.
	2022	Session Chair, Swarm Robotics, IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS).
	2018	Session Chair, Navigation Technology, IEEE/CSAA Guidance, Navigation and Control Conference.
	2016–2017	Chief Editor, Department Newsletter <i>Lift-Off</i> , IIT Bombay.
Peer Review		
Journals		Conferences
• <i>Science Advances</i>		• Robotics: Science and Systems
• <i>IEEE Transactions on Robotics</i>		• IEEE Conference on Decision and Control
• <i>IEEE Transactions on Automatic Control</i>		• IEEE International Conference on Robotics and Automation
• <i>The International Journal of Robotics Research</i>		• IEEE/RSJ International Conference on Intelligent Robots and Systems
• <i>Autonomous Robots</i>		• International Symposium on Distributed Autonomous Robotic Systems
• <i>IEEE Robotics and Automation Letters</i>		